

O'Reilly® | Live Course

• ENGINEERING FOR THE AI ERA

Introduction to AI Infrastructure

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- WHO I AM

Bryan Oliver

Principal Engineer

Platform Engineering, Thoughtworks AI Labs (TAILs), TW AI Factory, TW Radar Committee

International Speaker

Talks, workshops & keynotes — CNCF, KubeCon, QCon, Craft Conf, KubeDay, CDCon, and more.

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Effective Platform Engineering (Manning)

- ROADMAP

How today is laid out

1 Why GPUs Matter

From video-game frames to the engines behind every AI product. · 30 min

2 What is a “GPU Cloud”

The stack between your YAML and the silicon. · 30 min

3 Why GPU-to-GPU Speed Matters

When chips must talk fast, the network is the computer. · 30 min

4 Topology-Aware Scheduling

Placing work where the wires are fattest. · 40 min

5 Deploying & Delivering Workloads

Getting jobs onto the cluster and out the door. · 30 min

6 Hands-on with Kueue — capstone lab

Queue, quota, and schedule real workloads yourself. · 45 min

7 Synthesis & Wrap-up

Tie the day together; where to go next. · 25 min

Plus **two 5-min breaks** along the way.

Narrative: Why → Where → How (networking) → How (scheduling) → Do → Synthesize

- BEFORE WE START

Stay engaged

THROUGHOUT THE DAY

Chat

Think out loud, react, drop links.

Q&A widget

Park questions; we answer in batches.

Polls & pulse checks

Quick taps that steer our pace.

Whiteboard

We sketch a few ideas together.

Have ready

FOR THE HANDS-ON LAB

kubectl + helm

Installed and on your PATH.

A local cluster

kind or minikube is fine.

Code editor

Whatever you're comfortable in.

k9s or Lens

Optional, but handy for watching.

No GPUs required today — we [simulate the hardware](#) so everyone can follow along.